Problem Set 2

Graded

Student

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Total Points

100 / 100 pts

Question 1

Question 1

40 / 40 pts

- ✔ **– 0 pts** Correct
- ✔ **– 0 pts** Follow Homework format

– 1 pt Incorrect value used

– 1 pt Incorrect value used

– 1 pt Missing/Incorrect units

– 1 pt Missing/Incorrect units

– 1 pt Calculation error

– 1 pt Incorrect equation used

– 1 pt Incorrect equation used

– 1 pt Incorrect final answer

– 1 pt Incorrect final answer

– 10 pts Missing part C

– 40 pts Problem not attempted

Question 2

Question 2

40 / 40 pts

✔ **– 0 pts** Correct

✔ **– 0 pts** Follow homework format

– 20 pts Incorrect a)

– 10 pts Major error in a)

– 10 pts Incorrect b)

– 10 pts Incorrect c)

– 2 pts c) Error in strategy

– 5 pts c) 2 Errors in strategies

– 1 pt Incorrect/missing units

– 5 pts major error in energy balance

– 2 pts calculation error

Question 3

Question 3

20 / 20 pts

– 1 pt Missing the negative sign for the k

– 10 pts Incorrect b: claimed right side has larger heat flux

– 1 pt Select pages in the future for full credit.

✔ **– 0 pts** Perfect

– 10 pts Incorrect direction for a

– 5 pts Partially incorrect a: either direction for arrows is not fully correct or no arrows drawn at all

– 20 pts no submission

No questions assigned to the following page.

MAE 3240 HW2

2/10

Questions assigned to the following page: 1 and 2

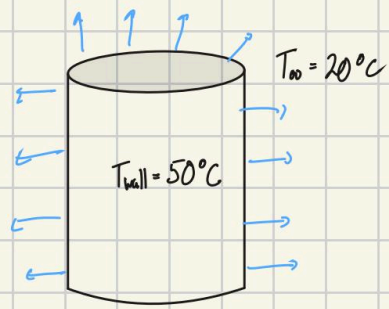
Known

- $R = 0.5m$, $H = 1m$, $T_{wall} = 50°C$
- $T_\infty = 20°C$, $h = 15 \frac{W}{m^2 \cdot K}$
- $L = .02m$, $k = 0.1 \frac{W}{mK}$

Find

- required electrical power no foam & foam
- impacts of adding foam on energy needed to insulate

Schematic



Assumptions

- Electrical energy in = heat out
- Bottom is well insulated
- Radiation is negligible

Analysis

a) $A_{tot} = A_{side} + A_{top} = 2\pi(0.5m)(1m) + \pi(0.5m)^2$
$= 3.93 m^2$

$Q = h \cdot A \cdot \Delta T = (15 \frac{W}{m^2 \cdot K}) \cdot (3.93 m^2)(30K) = 1.77 kW$

b) $R_{cond} = \frac{L}{k} = 0.2$ $R_{conv} = \frac{1}{h} = 0.067$

$R_{tot} = 0.267 \frac{m^2 \cdot K}{W}$

$q_{foam} = \frac{\Delta T}{R_{tot}} = 112.5 \, W/m^2 \cdot K$

$q_{tot} = q_{foam} \cdot A_{tot} = 442W$

c) $\frac{q_{tot}}{q_{no\,foam}} = 0.26$ → same ΔT produces smaller heat loss → less electrical input needed

2)

Known

- $L = 0.1m$, $k = 1 \frac{W}{m \cdot K}$
- $T_\infty = 20°C$, $h = 100 \frac{W}{m^2 \cdot K}$
- $q = 1000 \frac{W}{m^2}$
- $T_{max} = 200°C$

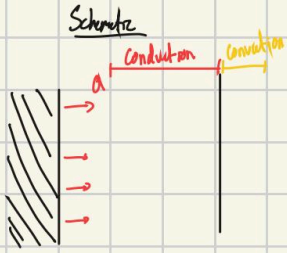
Find

- $T(0)$, $T(L)$
- q_{max} @ $T(0) < T_{max}$
- How to increase q_{max} @ $T(0) < T_{max}$

Assumptions

- no radiation, left side insulated
- Steady-state, constant k

Schematic



Analysis

a) $q_{convec} = h\Delta T = h[T(L) - T_\infty]$

↳ $T_L = T_\infty + \frac{q}{h} = 30°C$

$\frac{k \cdot \Delta T}{L} = q \rightarrow \frac{1(T_L - 20°C)}{0.1m} = 1000 \frac{W}{m^2 \cdot K} \rightarrow T_0 = 130°C$

b) $R_{tot} = \frac{L}{k} + \frac{1}{h} = 0.1 + 0.01 = 0.11$

$\Delta T_{max} = 180K$

$q_{tot} = \frac{\Delta T_{max}}{R_{tot}} = \frac{180K}{0.11} = 1.64 \frac{kW}{m^2}$

c) higher h, higher k, decrease L

higher heat flow — more conductive transfer — thinner layer for transfer

Question assigned to the following page: 3

Schematic + a)

Known

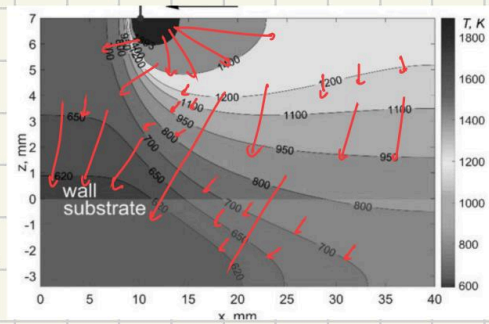
- given isotherm

Find

- heat flux vectors

Assumptions

- 2-D heat transfer in x & z directions



Analysis

$\vec{q} = -k \nabla T$

↳ moves from hot to cold

↳ perpendicular due to gradient

b) there is tighter contour spacing on the left side - so heat transfer gradient is greater